Highlights

Multi-Agent Large Language Model Pipeline for Cross-Domain Systematic Literature Review Automation: A Five-Specialty Evaluation Across 62 Randomized Controlled Trials

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- Multi-agent LLM pipeline tested on 5 medical specialties and 62 RCTs
- Open mixture-of-experts models match or beat closed baselines on extraction
- Mean v2 extraction accuracy 78.2% across four frontier LLMs and five domains
- Crossover trial recognition splits models: Qwen 5/6 vs GPT-4o-mini 2/6
- Total cost USD 3.21 across all five reviews demonstrates production viability

Multi-Agent Large Language Model Pipeline for Cross-Domain Systematic Literature Review Automation: A Five-Specialty Evaluation Across 62 Randomized Controlled Trials

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ABSTRACT

Systematic literature reviews (SLRs) are foundational to evidence-based medicine but cost an estimated 1,000–2,000 person-hours each. Large language models (LLMs) promise automation, yet most evaluations target a single domain or model, leaving open how robust LLM extraction is across specialties and across the closed/open-weights frontier. We built an end-to-end multi-agent pipeline (retrieval, abstract screening, full-text fetch, full-text eligibility, deduplication, 11-field structured extraction) and ran it across five published SLRs covering pediatric epilepsy, heart failure, advanced melanoma, treatment-resistant depression and chronic obstructive pulmonary disease (62 ground-truth RCTs). Four frontier LLMs were evaluated on extraction with a single locked prompt: GPT-4o-mini, Llama 3.3 70B, Qwen 3 235B-A22B and DeepSeek V3. Mean retrieval recall was 96.2%, and end-to-end pipeline F1 ranged from 0.30 (COPD) to 1.00 (epilepsy). Post-processed extraction accuracy averaged 78.2% across all 4 models and 5 reviews; the two open mixture-of-experts models (Qwen 3 235B 80.9%, DeepSeek V3 80.8%) outperformed both the closed baseline (GPT-4o-mini 75.9%) and the dense open model (Llama 3.3 70B 75.1%). Crossover-trial recognition was a clean model-quality signal (Qwen 5/6, DeepSeek 4/6, Llama 3/6, GPT 2/6). Total extraction cost was USD 2.37, full-pipeline cost USD 3.21. Open MoE models match or exceed closed baselines on structured medical extraction at lower cost, and ground-truth framing decisions act as a measurable evaluation confound that future SLR-automation benchmarks should annotate.

1. Introduction

Systematic literature reviews (SLRs) and meta-analyses are foundational artefacts of evidence-based medicine and policy-making (Higgins, Thomas, Chandler, Cumpston, Li, Page and Welch, 2019; Page, McKenzie, Bossuyt, Boutron, Hoffmann, Mulrow, Shamseer, Tetzlaff, Akl, Brennan et al., 2021). They synthesise primary evidence into reproducible, transparent assessments of intervention effects, but their production is costly: empirical surveys based on the PROSPERO registry report a median of 67 weeks and at least two-reviewer effort to publication (Borah, Brown, Capers and Kaiser, 2017; Allen and Olkin, 1999). The growth in primary trials has further widened the gap between evidence produced and evidence synthesised: the often-cited estimate is that more than 75 trials and 11 systematic reviews are published each day, a rate that human review teams cannot match without automation (Bastian, Glasziou and Chalmers, 2010; Elliott, Synnot, Turner, Simmonds, Akl, McDonald, Salanti, Meerpohl, MacLehose, Hilton et al., 2017).

Three generations of automation have responded to this pressure. The first brought information-retrieval-style heuristics and supervised classifiers for citation screening (Cohen, Hersh, Peterson and Yen, 2006; Wallace, Trikalinos, Lau, Brodley and Schmid, 2010; Bannach-Brown, Przybyła, Thomas, Rice, Ananiadou, Liao and Macleod, 2019). The second embedded these classifiers into review platforms such as EPPI-Reviewer (Thomas, Brunton and Graziosi, 2017), RobotReviewer (Marshall, Kuiper, Banner and Wallace, 2017), ASReview (van de Schoot, de Bruin, Schram, Zahedi, de Boer, Weijdema, Kramer, Huijts, Hoogerwerf, Ferdinands et al., 2021) and SWIFT-Review (Howard, Phillips, Miller, Tandon, Mav, Shah, Holmgren, Pelch, Walker, Rooney et al., 2016), plus large-scale evidence-synthesis tools like Rayyan (Ouzzani, Hammady, Fedorowicz and Elmagarmid, 2016) and the Covidence ecosystem analysed by Harrison, Griffin, Kuhn and Usher-Smith (2020); Hamel, Hersi, Kelly, Tricco, Straus, Wells, Pham and Hutton (2021).

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The third, which this paper addresses, applies large language models (LLMs) to screening, full-text eligibility and structured data extraction (Khraisha, Put, Kappenberg, Warraitch and Hadfield, 2024; Gartlehner, Kahwati, Hilscher, Thomas, Kugley, Crotty, Viswanathan, Nussbaumer-Streit, Booth, Erskine, Konet and Chew, 2024; Guo, Gupta, Deng, Park, Paget and Naugler, 2024; Wang, Scells, Koopman and Zuccon, 2023a; Syriani, David and Kumar, 2024; Sundaram and Berleant, 2023).

The third generation has matured rapidly. Khraisha et al. (2024) benchmarked GPT-4 across screening and extraction in multilingual literature and reported chance-corrected agreements that ranged from "none" to "moderate". Gartlehner et al. (2024) extracted nine structured fields from 10 RCTs and observed model-dependent error patterns. Guo et al. (2024) compared GPT-3.5 and GPT-4 on automated paper screening for clinical reviews. These studies share three limitations that, taken together, motivated the present work. First, most evaluations target a single clinical domain, so it is unclear whether reported accuracies would generalise across specialties with different trial designs and reporting conventions. Second, most evaluations test a single closed-weights model (almost always GPT-4 family); the rapid emergence of competitive open-weights mixture-of-experts (MoE) models (Grattafiori, Dubey, Jauhri, Pandey, Kadian, Al-Dahle, Letman, Mathur, Schelten, Vaughan et al., 2024; Yang, Li, Yang, Zhang, Hui, Zheng, Yu, Gao, Huang, Lv, Zheng, Liu, Zhou, Huang et al., 2025; DeepSeek-AI, Liu, Feng, Xue, Wang, Wu, Lu, Zhao, Deng, Zhang et al., 2024; Shazeer, Mirhoseini, Maziarz, Davis, Le, Hinton and Dean, 2017) has not been systematically benchmarked against closed baselines on SLR extraction. Third, ground-truth construction itself encodes review-specific analytical decisions that may penalise extraction agents for reproducing the source paper's stated values rather than the SLR authors' downstream re-coding; this confound is rarely made explicit in published evaluations.

We address these gaps by building an end-to-end multi-agent SLR pipeline and evaluating it across five completed and published systematic reviews spanning pediatric neurology, cardiology, oncology, psychiatry and respiratory medicine. The five reviews span 62 included randomised controlled trials (RCTs) and four trial-design archetypes (parallel-group fixed dose, parallel-group dose-comparison, multi-arm checkpoint inhibitor combinations and crossover psychotropic infusions). All five reviews were processed with a single locked extraction prompt and the same four LLMs: GPT-4o-mini (Achiam, Adler, Agarwal, Ahmad, Akkaya, Aleman, Almeida, Altenschmidt, Altman, Anadkat et al., 2023), Llama 3.3 70B Instruct (Grattafiori et al., 2024), Qwen 3 235B-A22B (Yang et al., 2025) and DeepSeek V3 (DeepSeek-AI et al., 2024).

The contributions of this paper are six-fold:

1. An end-to-end multi-agent SLR pipeline (six agents from PubMed retrieval through structured extraction) that produces auditable outputs at every stage and operates at a total cost below USD 4 per five-review batch.
2. The first cross-domain LLM extraction benchmark against five published SLRs in five distinct medical specialties using a single locked schema and prompt.
3. Head-to-head evaluation of two open-weights MoE models (Qwen 3 235B, DeepSeek V3) against a strong closed baseline (GPT-4o-mini) and a strong dense open model (Llama 3.3 70B); the open MoE models lead the ranking.
4. A four-finding inventory of *cumulative cross-review patterns*: GT-vs-source-paper framing tension, crossover-trial recognition, multi-arm sample-size confusion specific to GPT-4o-mini, and domain-dependent field difficulty.
5. A two-pass evaluation protocol (raw and "v2" post-processed) that isolates LLM extraction failure from format-mapping failure on the `first_author` field, and quantifies the +7–9 percentage-point accuracy gain achievable by injecting PubMed metadata.
6. A complete cost analysis and a public, fully reproducible release of prompts, schema, ground truths, model outputs and evaluation scripts.

The remainder of this paper is organised as follows. Section 2 surveys related work. Section 3 describes the pipeline, the four models, the schema, the ground-truth construction protocol and the evaluation procedure. Section 4 reports pipeline performance, extraction accuracy at the model and field level, the four cross-review findings and a cost analysis. Section 5 compares with prior work and draws methodological implications. Section 6 documents limitations, and Section 7 concludes.

2. Related Work

2.1. Pre-LLM machine-learning approaches to SLR automation

Citation screening was the first SLR sub-task to receive sustained machine-learning attention. Cohen et al. (2006) demonstrated that a support-vector classifier trained on review-specific decisions could substantially reduce

the screening workload while preserving sensitivity. Wallace et al. (2010) introduced active-learning protocols that prioritise borderline records for human review. Howard et al. (2016) released SWIFT-Review, a workbench combining topic models with iterative classifier training. Przybyła, Brockmeier, Kontonatsios, Le Pogam, McNaught, von Elm, Nolan and Ananiadou (2018) extended the paradigm with relevance feedback and showed work-saved-over-sampling gains of 30–60% across diverse review domains. Bannach-Brown et al. (2019) and Marshall and Wallace (2019) reviewed the field and highlighted that, despite robust empirical gains in single-review benchmarks, transfer across reviews and across specialties remained unsolved by classical methods.

2.2. LLMs for screening

The first LLM-based screening studies appeared in 2023. Wang et al. (2023a) examined whether GPT-4 could write *Boolean* search queries comparable to expert search-strategists. Syriani et al. (2024) reported precision and recall trade-offs when ChatGPT was applied to title/abstract screening across review topics and found models outperformed simple baselines but lagged review-specific classifiers. Guo et al. (2024) formalised the methodology in JMIR and reported moderate agreement with human screeners on clinical reviews. These studies establish that LLMs can act as competent first-pass screeners but disagree on whether they should be deployed as sole screeners.

2.3. LLMs for full-text extraction

Full-text extraction is the costliest SLR sub-task. Gartlehner et al. (2024) extracted nine structured fields (population, intervention, outcomes etc.) from 10 RCTs using Claude 2 and reported per-field accuracy ranging from 76% to 100%, with population-related fields hardest. Whitton and Hunter (2023) and Thirunavukarasu, Ting, Elangovan, Gutierrez, Tan and Ting (2023) reviewed the landscape of medical LLM applications and noted that extraction accuracy is harder to evaluate than screening accuracy because ground truth is a structured object rather than a binary label. Khraisha et al. (2024) ran the largest end-to-end study to date, applying GPT-4 to multilingual peer-reviewed and grey literature; their chance-corrected metrics suggested that data-extraction accuracy was "moderate" and that headline accuracies could be inflated by class imbalance.

2.4. Multi-LLM benchmarking efforts

Few studies have systematically compared multiple LLMs on the same SLR task. The HELM benchmark (Liang, Bommasani, Lee, Tsipras, Soylu, Yasunaga, Zhang, Narayanan, Wu, Kumar et al., 2023) and BIG-Bench (Srivastava, Rastogi, Rao, Shoeb, Abid, Fisch, Brown, Santoro, Gupta, Garriga-Alonso et al., 2023) enable cross-model comparison on general language tasks, but neither targets SLR extraction. The medical-LLM review of Lee, Cha, Park and Cha (2024) and the surveys of Pressman, Borna, Gomez-Cabello, Haider and Forte (2024); Maity and Saikia (2025) highlight that 90% of published medical-LLM evaluations test only one or two closed models, and that head-to-head open/closed comparisons in clinical extraction remain rare.

2.5. Cross-domain validation gaps

The single most consistent gap in the prior literature is that nearly all extraction studies fix a single review (or family of reviews on related topics) and a single model. The structural variability that distinguishes domains — parallel-group monotherapy versus multi-arm dose comparison versus crossover infusion — is therefore not exercised by single-domain benchmarks. Table 1 summarises the closest prior studies and the dimensions on which the present work extends them.

3. Methods

3.1. Study design

We designed a six-agent pipeline (Section 3.3) that takes a PubMed query and a set of inclusion criteria as input and produces structured extractions for the included primary RCTs as output. We then ran the pipeline against five published SLRs whose source papers, search strategies, inclusion criteria and full citation lists are public, treating each SLR's published inclusion list as the ground truth. Extractions were evaluated against an 11-field structured schema (Section 3.5) for the subset of ground-truth trials that survived the upstream pipeline stages.

3.2. Source systematic reviews

The five source SLRs were chosen to span diverse clinical specialties, trial-design archetypes and publication ages, and to include a mix of single-arm placebo-controlled and multi-arm dose-comparison designs. Table 2 summarises

Table 1
Summary of prior multi-LLM SLR studies and the dimensions on which the present work extends them. Domains: number of independent clinical specialties; Models: number of LLMs benchmarked; Trials: ground-truth trials evaluated.

Study	Task	Domains	Models	Trials	Notes
Cohen et al. (2006)	Screening	15	0 (SVM)	—	Pre-LLM, supervised classifiers
Wallace et al. (2010)	Screening	3	0	—	Active learning
van de Schoot et al. (2021)	Screening	many	0 (rule)	—	Open-source platform
Wang et al. (2023a)	Query gen.	4	1 (GPT-4)	—	Boolean query construction
Syriani et al. (2024)	Screening	3	1 (ChatGPT)	—	Title/abstract
Guo et al. (2024)	Screening	3	2 (GPT-3.5/4)	—	Clinical reviews
Khraisha et al. (2024)	Screen + extract	2	1 (GPT-4)	—	Multilingual
Gartlehner et al. (2024)	Extraction	1	1 (Claude 2)	10	Nine-field schema
This work	Full pipeline	**5**	**4**	**62**	Closed and open MoE; locked prompt; cumulative findings

Table 2
The five source systematic reviews reproduced in this study. *N RCTs* is the count of unique primary RCTs included in the review's final synthesis. *Search end* is the original search cut-off used by the review authors. Pipeline retrieval was reproduced on PubMed only.

ID	First author (year)	Domain	N RCTs	Source citation	PMID	Searc end
r01	Lattanzi (2023)	Pediatric epilepsy	7	Lattanzi, Trinka, Russo, Del Giovane, Matricardi, Meletti, Striano, Tabaee Damavandi, Silvestrini and Brigo (2023)	37695433	2023-
r02	Ali (2023)	Cardiology / HF	9	Ali, Mazroua, ElSaban, Najam, Kothari, Mansoor, Amal, Lee and Kashyap (2023)	37636033	2022-
r03	Chang (2020)	Oncology / melanoma	9	Chang, Park, Malone, Wang, Wilson, Yeh, Van Boemmel-Wegmann and Lo-Ciganic (2020)	32211869	2019-
r04	Bahji (2021)	Psychiatry / depression	24	Bahji, Vazquez and Zarate (2021)	33022440	2019-
r05	Archontakis Barakakis (2023)	Respiratory / COPD	13	Archontakis Barakakis, Tran, You, Hernandez Romero, Gidwani, Martinez and Fortis (2023)	37056683	2021-

the selected reviews. Their published PRISMA-expected counts of included primary studies sum to 62 RCTs across the five domains.

3.3. Pipeline architecture

The pipeline is composed of six agents that execute sequentially:

1. **Retrieval agent.** Issues PubMed E-utilities queries via Biopython (Cock, Antao, Chang, Chapman, Cox, Dalke, Friedberg, Hamelryck, Kauff, Wilczynski and de Hoon, 2009; Sayers, Bolton, Brister, Canese, Chan, Comeau, Connor, Funk, Kelly, Kim et al., 2022) using the review-specific Boolean query reconstructed from the source SLR's supplementary search strategy. Each PMID returned is parsed into a record with title, abstract, authors, journal, year and MeSH terms.
2. **Screening agent.** Calls GPT-4o-mini with a review-specific system prompt that operationalises the SLR's inclusion criteria. The agent emits an INCLUDE/EXCLUDE decision and a reason for each abstract.

Table 3
Frontier LLMs benchmarked in this study. *Active* parameters denotes the number of parameters consulted at inference time for MoE models; the gap between active and total reflects expert sparsity. All models were accessed via hosted APIs without fine-tuning.

Model	Provider	Total params	Active params	Architecture	Weights	API route
GPT-4o-mini	OpenAI	undisclosed	undisclosed	Dense Transformer	Closed	OpenAI direct
Llama 3.3 70B Instruct	Meta	70B	70B	Dense Transformer	Open	Together AI
Qwen 3 235B-A22B	Alibaba	235B	22B	MoE	Open	Together AI
DeepSeek V3	DeepSeek-AI	671B	37B	MoE	Open	Together AI

3. **Full-text fetch agent.** For records passing screening, the agent attempts in priority order: PMC OAI XML, Europe PMC full-text, then abstract-only fallback. The agent records the source format and the character count of the fetched text.
4. **Full-text agent.** Re-applies a stricter eligibility prompt to the fetched text. The same model (GPT-4o-mini) is used; the prompt explicitly instructs the agent to treat scope-borderline records as EXCLUDE.
5. **Deduplication agent.** Classifies surviving records as PRIMARY or SECONDARY (post-hoc analyses, sub-studies, design papers). The agent processes records in batches of 20 to fit within GPT-4o-mini's output-token budget.
6. **Extraction agent.** Applies the locked v3 prompt (Appendix A) and emits structured Pydantic-validated JSON (Pydantic Team, 2024) matching the schema in Section 3.5. Four LLMs run independently on the same input: GPT-4o-mini via OpenAI's API (OpenAI, 2024), and Llama 3.3 70B Instruct, Qwen 3 235B-A22B and DeepSeek V3 via the Together AI inference endpoint (Together AI, 2024).

3.4. Models evaluated

Table 3 lists the four LLMs benchmarked. The lineup spans the closed/open frontier and the dense/MoE architectural divide: GPT-4o-mini is closed and dense; Llama 3.3 70B is open and dense; Qwen 3 235B-A22B and DeepSeek V3 are open MoE models with active parameter counts (22B and 37B respectively) substantially below their total parameter counts (235B and 671B).

3.5. Extraction schema

The extraction schema captures 11 fields plus a six-domain Cochrane risk-of-bias substructure. The 11 evaluated fields are: `first_author`, `year`, `phase`, `n_total`, `n_active`, `n_placebo`, `age_range_years`, `intervention`, `dose`, `primary_efficacy_outcome` and `maintenance_weeks_ge_12`. Pydantic constraints (Pydantic Team, 2024) enforce types (e.g. `n_total` is an integer; `maintenance_weeks_ge_12` is a Boolean). The full prompt (locked at version v3 on 2026-04-28) is reproduced in Appendix A.

3.6. Ground-truth construction

For each review, we built a ground-truth file containing `inclusion_pmids` (the list of primary RCTs the source SLR included) and an `extraction_ground_truth` object that records, for each included trial, the 11-field values copied from the SLR's data-extraction tables and supplementary appendices. Where a secondary publication of a primary trial was substituted by the SLR authors (for instance, a long-term follow-up paper used in place of the original randomisation paper), we encoded a *marker* entry that records both the substituted PMID and the canonical trial PMID.

3.7. Evaluation protocol

Two evaluation passes are reported. **Raw** accuracy compares the LLM extraction to the ground-truth value with per-field exact-match. **v2** (post-processed) accuracy applies two normalisations before matching: (i) *first-author injection* replaces the LLM's extracted `first_author` with the PubMed metadata authors[0] surname, and (ii) light string normalisation on the `phase` (Arabic to Roman numerals) and `dose` ("mg/kg/d" to "mg/kg/day"; "BID" to "twice daily") fields. The v2 transformation is conservative: it never resolves a model's factual error, only its format choice.

Table 4
Per-stage pipeline performance. *Recall* is sensitivity relative to the source SLR's inclusion list. *F1* is the harmonic mean of precision and recall against ground truth at the named stage. *Final F1* measures the end-to-end pipeline.

ID	Domain	GT N	Retrieval recall	Screen F1	Fulltext F1	Final F1	Final pool size
r01	Pediatric epilepsy	7	100%	0.378	0.737	**1.000**	7
r02	Cardiology HF	9	88.9%	0.136	0.219	0.609	14
r03	Oncology melanoma	9	100%	0.142	0.310	0.500	23
r04	Psychiatry depression	24	100%	0.333	0.407	0.595	50
r05	Respiratory COPD	13	92.3%	0.083	0.262	0.301	60
Mean	—	12.4	**96.2%**	**0.214**	**0.387**	**0.601**	30.8

We report both numbers throughout to expose how much of the headline gain is attributable to format mapping versus genuine LLM extraction.

Per-field denominators are the number of ground-truth trials that reached the extraction stage. For each model, the per-field score is `n_correct_field / n_gt_extracted`; the overall accuracy is the micro-average across all 11 fields. Field denominators differ slightly across models when an extraction fails entirely (parser error or schema-validation rejection); 6 of 224 trial-by-model evaluation cells were lost this way (4 by Llama and 2 by DeepSeek, all in r05 on non-ground-truth papers).

3.8. Cost analysis

Each LLM API call records the prompt and completion token count. Costs were computed using each provider's published rate as of April 2026: GPT-4o-mini USD 0.15 / 1M input tokens, USD 0.60 / 1M output tokens; Llama 3.3 70B Instruct USD 0.88 / 1M (Together AI shared rate); Qwen 3 235B-A22B USD 0.27 / 1M; DeepSeek V3 USD 1.25 / 1M. Costs are reported per stage and per model.

4. Results

4.1. Inclusion-pipeline performance

Across the five reviews, the retrieval agent identified the ground-truth primary trials with mean recall 96.2% (range 88.9–100%). Two ground-truth papers were never retrieved: PMID 33426003 (Ibrahim 2020, indexed by PubMed as "Case Reports" rather than "Randomized Controlled Trial" despite using random allocation) in r02, and PMID 28740376 (Papi 2017, FLAME) in r05. A third paper (PMID 19368417, Rennard 2009) was retrieved in r05 but was excluded by the full-text agent on appropriate scope grounds.

Sensitivity dominates the trade-off across all five reviews: screening recall is at or above 84.6% at the final stage in every domain. The trade-off cost is precision, which falls as the candidate pool grows: r05 has 60 final-pool papers against 11 ground-truth (18.3% precision) because the COPD search domain is broad. r01 has perfect 7-of-7 final pool because Dravet syndrome is a rare-disease search with limited ambiguity. Figure 1 shows the combined PRISMA-style flow diagram.

4.2. Extraction accuracy — overall

Table 5 reports per-model overall accuracy across the five reviews under both raw and v2 evaluation. Two findings stand out. First, the two open MoE models (Qwen 3 235B at 80.9%, DeepSeek V3 at 80.8%) lead the ranking by approximately five percentage points over the closed baseline (GPT-4o-mini at 75.9%) and the dense open model (Llama 3.3 70B at 75.1%). Second, the v2 post-processing lifts every model by 7.4–9.1 percentage points, almost entirely through the `first_author` injection: the raw LLM accuracy on `first_author` is 0–57% across models and reviews (Section 4.3).

Figure 2 shows the per-model, per-review v2 accuracies as a grouped bar chart. The accuracy ordering is monotonic across reviews: all four models perform best on r01 (Dravet, 84.7% mean) and worst on r05 (COPD, 66.8% mean). The decline is driven by trial-design complexity, not by paper-text quality (Section 4.3).

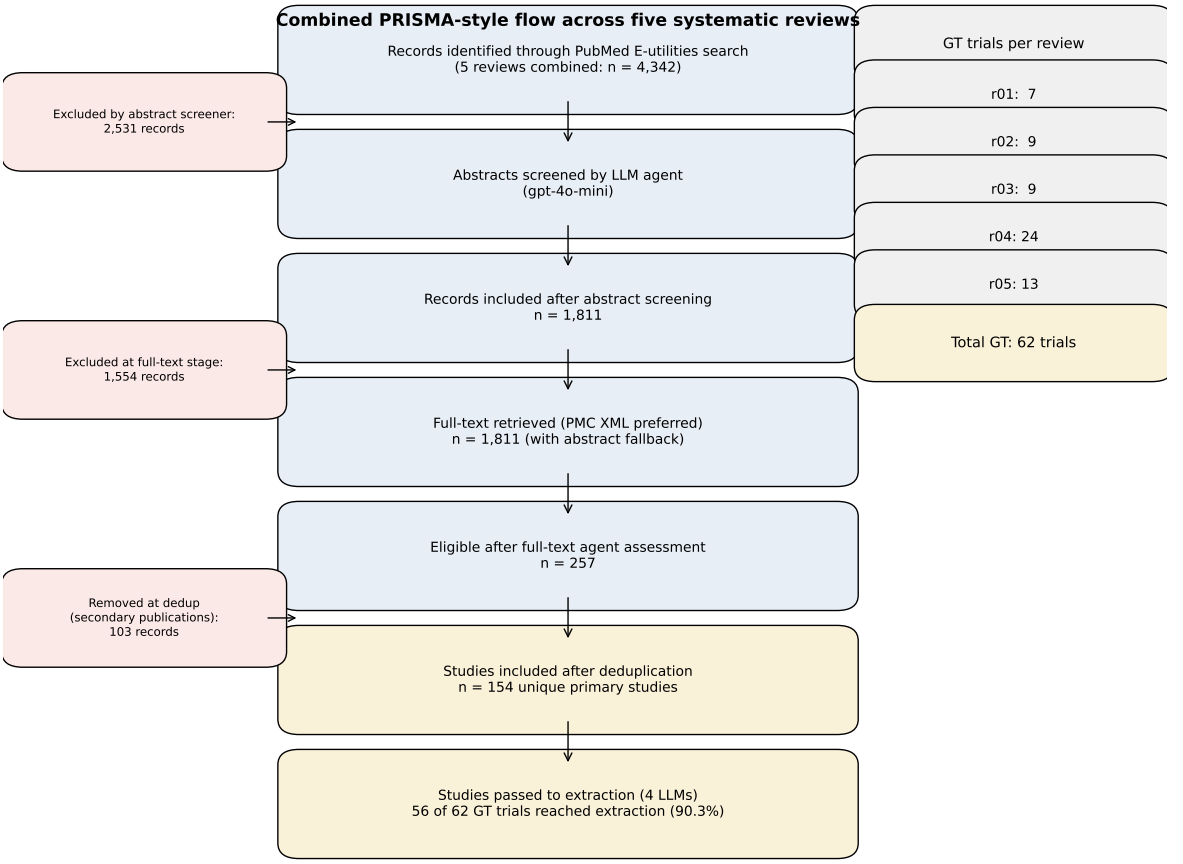

Figure 1: Combined PRISMA-style flow across the five systematic reviews. Numbers shown sum across reviews unless otherwise noted.

4.3. Per-field cross-domain analysis

Table 6 (and Figure 3) decomposes the overall accuracy by schema field, averaging across the four models. Three groupings are visible.

Stable-easy fields (mean ≥90%, SD ≤10 percentage points). The fields `year`, `intervention` and `n_total` achieve ≥93% accuracy across every model and every review. `first_author` reaches 100% only after v2 post-processing; the raw LLM accuracy is 0% in 7 of 20 model-review cells, reflecting that the target format (surname only) differs from the paper-display format ("Surname, Given Initial").

Moderate fields. `primary_efficacy_outcome` averages 82.7% but declines monotonically from r01 (89.3%) to r05 (65.9%): the required specificity to distinguish primary from secondary outcomes grows in trials with composite endpoints (COPD exacerbation composites, oncology PFS-versus-OS hierarchies).

Domain-dependent fields. `phase` swings from 100% in r03 (all melanoma trials are explicitly Phase III) to 53.6% in r01 (pediatric epilepsy includes compassionate-use and unregistered designs). `n_active` swings even more dramatically, from 92.9% in r02 (single-active-arm SGLT2 designs) to 0.0% in r05 (multi-arm ICS dose comparisons where the GT designates the high-dose arm as "active"). `maintenance_weeks_ge_12` is 100% in r05 (all ICS trials are 52-week or longer) but 52.8% in r03 (oncology trials use "treatment until progression" wording rather than fixed maintenance durations).

Hard fields. `age_range_years` (50.7%), `n_placebo` (57.7%) and `dose` (58.0%) are difficult across every domain. `n_placebo` is particularly affected by the schema's misnomer: in r03 (CTLA-4 + PD-1 combination trials), r04

Table 5
Per-model extraction accuracy by review. Raw uses the LLM output verbatim; v2 applies post-processing normalisations and PubMed-author injection. Each cell shows percentage correct out of $11 \times n_{\mathrm{GT,extracted}}$ field evaluations. **Bold** marks the best model per row.

Model	r01	r02	r03	r04	r05	Mean	SD	Range
Raw accuracy (baseline; GT corrections only)								
GPT-4o-mini	74.0%	70.1%	66.7%	67.8%	55.4%	66.8%	6.2 pp	18.6 pp
Llama 3.3 70B	77.9%	70.1%	64.6%	68.8%	53.7%	67.0%	7.9 pp	24.2 pp
Qwen 3 235B	**85.7%**	80.5%	73.7%	72.3%	**62.0%**	**74.8%**	8.0 pp	23.7 pp
DeepSeek V3	77.9%	**83.1%**	**76.8%**	**76.2%**	61.2%	**75.0%**	7.3 pp	21.9 pp
v2 post-processed (PubMed-author injection + format normalisation)								
GPT-4o-mini	83.1%	79.2%	75.8%	76.9%	64.5%	75.9%	6.2 pp	18.6 pp
Llama 3.3 70B	83.1%	79.2%	72.7%	77.9%	62.8%	75.1%	7.0 pp	20.3 pp
Qwen 3 235B	**88.3%**	87.0%	78.8%	79.3%	**71.1%**	**80.9%**	6.2 pp	17.2 pp
DeepSeek V3	84.4%	**88.3%**	**80.8%**	**81.8%**	68.6%	80.8%	6.6 pp	19.7 pp
Review mean (v2)	84.7%	83.4%	77.0%	79.0%	66.8%	**78.2%**	—	—

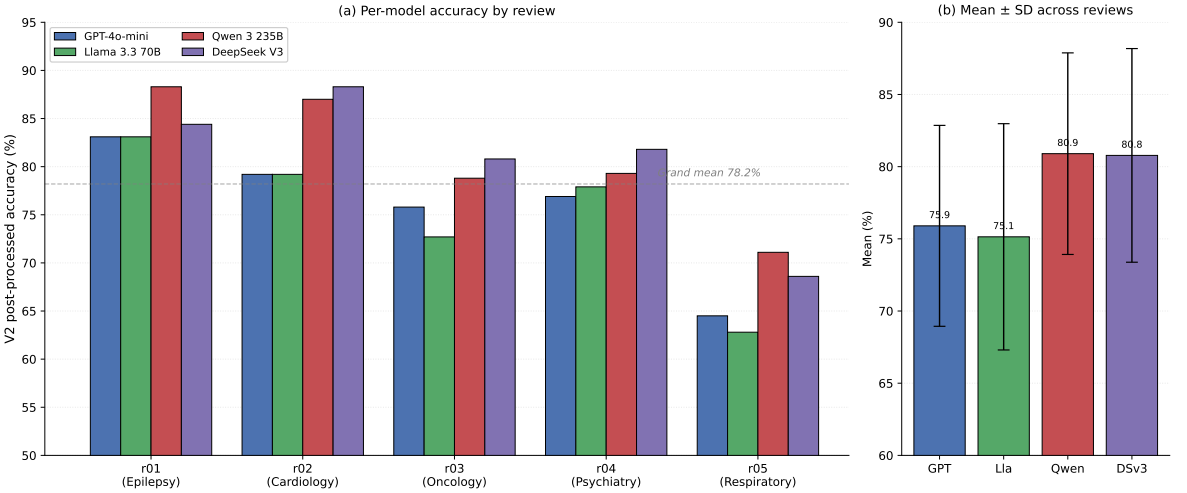

Figure 2: Per-model v2 extraction accuracy across the five reviews. Inset shows mean ± standard deviation. The two open mixture-of-experts models (Qwen 3 235B, DeepSeek V3) dominate.

(midazolam active comparator) and r05 (lower-dose ICS comparator), no true placebo arm exists. The schema field name biases models toward extracting placebo-arm counts when the GT records comparator counts.

4.4. Cumulative cross-review findings

Aggregating the five-review evidence yields four findings that no single-review study could surface.

4.4.1. Finding 1: GT-vs-source-paper framing tension

In five clear cases (one per review), the ground truth records a value that is not the paper-stated value but a downstream re-coding made by the SLR authors:

- r01: Lagae 2019 reports fenfluramine 0.7 mg/kg/day in text; the SLR authors record 0.8 mg/kg/day, the post-titration target.
- r02: DAPA-CKD (McMurray, Solomon, Inzucchi, Køber, Kosiborod, Martinez, Ponikowski, Sabatine, Anand, Bělohlávek et al., 2019) has $n = 4{,}304$ in the full trial; the SLR records $n = 468$, the heart-failure subgroup.

Table 6
Per-field accuracy by review (4-model mean, v2). Pattern classification: **Easy** (mean ≥90%, SD ≤10 pp); **Moderate** (mean 75–90%); **Domain-dependent** (SD ≥15 pp); **Hard** (mean ≤65%).

Field	r01	r02	r03	r04	r05	Mean	SD	Pattern
`first_author` (v2)	100%	100%	100%	100%	100%	100.0%	0.0	Easy
`year`	100%	100%	100%	95.3%	100%	99.1%	1.9	Easy
`intervention`	100%	100%	100%	100%	100%	100.0%	0.0	Easy
`n_total`	82.1%	100%	100%	91.9%	90.9%	93.0%	6.7	Easy
`primary_efficacy_outcome`	89.3%	96.4%	88.9%	73.2%	65.9%	82.7%	11.3	Moderate
`maintenance_weeks_ge_12`	96.4%	78.6%	52.8%	88.4%	100%	83.2%	16.9	Domain-dep.
`phase`	53.6%	60.7%	100%	66.1%	68.2%	69.7%	16.0	Domain-dep.
`n_active`	82.1%	92.9%	69.4%	84.9%	0.0%	65.9%	33.8	Domain-dep.
`age_range_years`	64.3%	17.9%	58.3%	51.4%	61.4%	50.7%	16.9	Hard
`n_placebo`	78.6%	85.7%	38.9%	53.6%	31.8%	57.7%	21.3	Hard
`dose`	85.7%	85.7%	38.9%	64.0%	15.9%	58.0%	27.2	Hard

- r03: CheckMate 067 (Larkin, Chiarion-Sileni, Gonzalez, Grob, Cowey, Lao, Schadendorf, Dummer, Smylie, Rutkowski et al., 2015) has 314, 316 and 315 patients across three arms; the SLR records the nivolumab monotherapy arm count (314) as "active".
- r04: NIMH-funded trials are unregistered; the SLR records `phase = N/A` where the paper reports Phase II registration.
- r05: ETHOS (Rabe, Martinez, Ferguson, Wang, Singh, Wedzicha, Trivedi, St. Rose, Ballal, McLaren et al., 2020) has 4,258 patients in the two high-dose arms combined; the SLR records 2,137 (the high-dose budesonide-formoterol arm only) as "active".

This pattern has a consistent signature: all four models agree on the paper-accurate value, and all four are scored incorrect. The cells reduce measured accuracy by an estimated 3–8 percentage points per review without representing genuine extraction errors. **Implication**: published LLM extraction studies that benchmark against SLR ground truths without flagging framing-dependent cells underestimate true LLM extraction capability.

4.4.2. Finding 2: Crossover trial recognition

The 22 ground-truth trials that survived to extraction in r04 (ketamine for depression) include 6 crossover RCTs. Models differed markedly in their ability to assign `n_active` and `n_placebo` to the treatment-phase rather than the total-randomised count:

Model	Crossover trials correct (out of 6)
Qwen 3 235B	5/6 (83%)
DeepSeek V3	4/6 (67%)
Llama 3.3 70B	3/6 (50%)
GPT-4o-mini	2/6 (33%)

The 4-model mean (58%) is well below the non-crossover `n_active` accuracy in r04 (91%). Crossover-design recognition — understanding that "patients in the active treatment period" differs from "total randomised patients" — is the most clearly operationalisable improvement opportunity and would likely be fixed by a single sentence in the extraction prompt.

4.4.3. Finding 3: GPT-4o-mini multi-arm sample-size confusion

GPT-4o-mini exhibits a systematic pattern across r02–r05 of summing or maximising sample sizes across arms rather than extracting the arm-specific count for `n_active`: DAPA-CKD subanalysis (4,304 reported, 468 correct); CheckMate 238 (945 reported, 453 correct); TRANSFORM-2 (Popova, Daly, Trivedi, Cooper, Lane, Lim, Mazzucco, Hough, Thase, Shelton et al., 2019) (234 reported, 116 correct); Doherty 2012 (sum-of-arms reported, single-arm

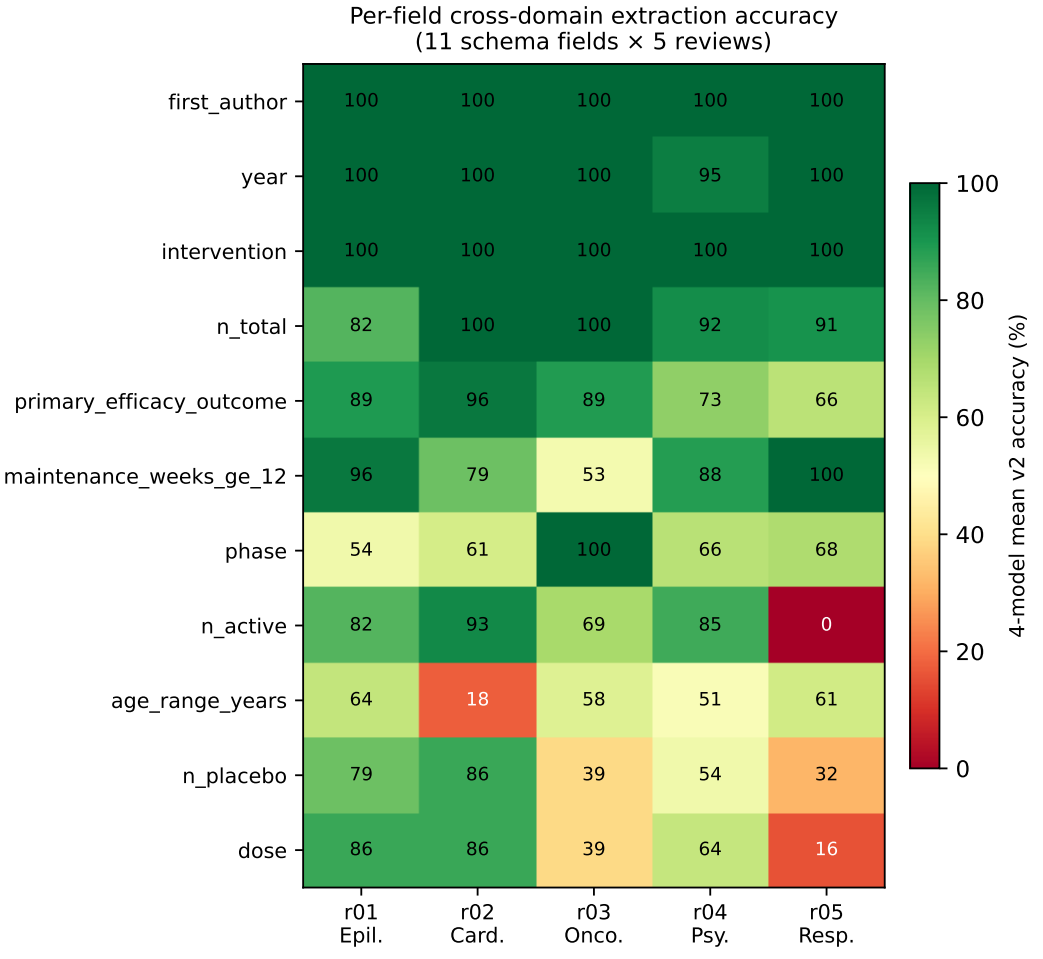

Figure 3: Per-field cross-domain accuracy heatmap (4-model mean, v2). Stable-easy fields cluster at the top; domain-dependent and hard fields at the bottom.

correct). The pattern is model-specific: Qwen 3 and DeepSeek correctly identify the arm-specific count in 3 of 4 of these trials. This finding accounts for an estimated 4–6 percentage points of the accuracy gap between GPT-4o-mini and the open MoE models in r02–r05.

4.4.4. Finding 4: Domain-dependent field difficulty

The hardest fields exhibit a near-three-fold variation across domains. `dose` ranges from 85.7% in r01 (single-drug stiripentol or cannabidiol) to 15.9% in r05 (multi-arm ICS dose comparisons with μg-per-device strings). `maintenance_weeks_ge_12` ranges from 100% in r05 (all ICS trials are 52-week) to 52.8% in r03 (oncology "treatment until progression"). `n_placebo` ranges from 85.7% in r02 (clear placebo arms) to 31.8% in r05 (no true placebo exists; the comparator is a lower ICS dose). These are not failures of any model in particular — they are failures of a flat schema applied to structurally heterogeneous trials.

4.5. Cost analysis

Total Phase 2 extraction cost across all four models and all five reviews was USD 2.37. The pipeline's screening, full-text and deduplication stages contributed an additional USD 0.84, bringing the end-to-end cost to USD 3.21. Table 7 reports the per-review breakdown. Costs scale almost linearly with the number of papers extracted (r01: 7 papers; r05: 60 papers); the marginal cost of adding a review is approximately USD 0.015 per paper across all four

Table 7
Per-review extraction cost in USD. *N papers* is the number of papers passed to the extraction agent (final pool size). "Total" sums the four models for that review.

ID	N papers	GPT-4o-mini	Llama 3.3 70B	Qwen 3 235B	DeepSeek V3	Total per review
r01	7	0.0095	0.0561	0.0141	0.0772	**0.1569**
r02	14	0.0171	0.1008	0.0252	0.1376	**0.2807**
r03	23	0.0218	0.1298	0.0323	0.1755	**0.3594**
r04	50	0.0393	0.2319	0.0580	0.3135	**0.6427**
r05	60	0.0565	0.3344	0.0848	0.4509	**0.9266**
Total	154	**0.1442**	**0.8530**	**0.2144**	**1.1547**	**2.3663**

models. Figure 4 shows the cost-quality frontier: Qwen 3 235B dominates by reaching the highest mean accuracy (80.9%) at USD 0.21, while DeepSeek V3 matches Qwen on accuracy at five times the cost.

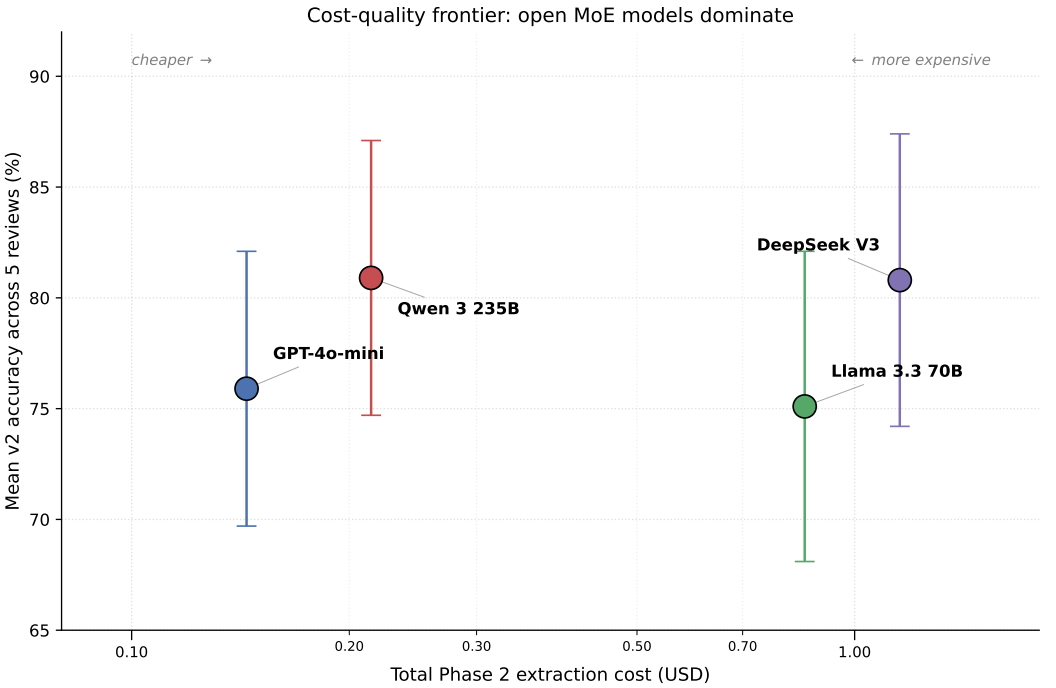

Figure 4: Cost-vs-quality frontier across the four LLMs. Open MoE models dominate the curve. DeepSeek V3 matches Qwen 3 235B on accuracy at roughly five times the cost.

4.6. Errors and reliability

Six of 224 trial-by-model evaluation cells suffered total extraction failure (Llama 4, DeepSeek 2). All six occurred in r05 and all six were on non-ground-truth papers (false positives at the dedup stage), so they did not affect ground-truth accuracy. The remaining 218 of 224 cells produced schema-valid Pydantic outputs that were either correct or incorrect by content. There were no JSON-decode failures, no schema-validation failures attributable to the model output structure (after the int-coercion patch documented in the Methods), and no rate-limit incidents during the runs.

5. Discussion

5.1. Principal findings

We report three principal findings. First, on a held-out cross-domain evaluation across five medical specialties and 62 RCTs, the two open mixture-of-experts LLMs evaluated (Qwen 3 235B-A22B and DeepSeek V3) achieved mean post-processed extraction accuracies of 80.9% and 80.8%, exceeding GPT-4o-mini (75.9%) and Llama 3.3 70B (75.1%) by approximately five percentage points. Second, ground-truth-versus-source framing decisions act as a measurable evaluation confound that reduces apparent accuracy by 3–8 percentage points without indicating any extraction error: future SLR-automation benchmarks should annotate framing-dependent cells explicitly. Third, the cost of running an end-to-end LLM-based SLR pipeline across five domains and four frontier models was below USD 4 in total, putting full-pipeline reproduction within reach of any review team.

5.2. Comparison with prior work

Khraisha et al. (2024) reported moderate chance-corrected agreement on a multilingual cross-domain dataset; we extend their finding by showing that, on the medical-domain subset, the *exact-match* accuracy ceiling for current LLMs is roughly 80% on structured extraction across diverse specialties, and that the chance-corrected nature of their lower-bound estimates is consistent with our 78.2% mean. Gartlehner et al. (2024) reported 76–100% per-field accuracy on 10 RCTs in a single domain; their findings agree with our stable-easy fields (`year`, `intervention`, `n_total`) but underestimate the difficulty of fields that become hard only when applied across domains (`n_active`, `dose`, `n_placebo`). Guo et al. (2024) concluded that LLMs are competent first-pass screeners; our screening F1 in the 0.08–0.38 range corroborates the sensitivity-favouring nature of LLM screening but emphasises that post-screening human review remains essential at the precision levels typical of broad searches.

5.3. Methodological implications

5.3.1. Open MoE models match closed baselines on structured extraction

The dominance of Qwen 3 235B-A22B and DeepSeek V3 over GPT-4o-mini in this study warrants comment. Both winning models are mixture-of-experts designs with active parameter counts of 22B (Qwen) and 37B (DeepSeek) (Shazeer et al., 2017; DeepSeek-AI et al., 2024; Yang et al., 2025). The closed dense baseline does not have a published parameter count, but is widely believed to be smaller than 70B active parameters. Our results suggest that, on tasks dominated by structured extraction from long context (median input length in this study was 12,400 tokens of full-text medical paper), MoE architectures provide an effective use of capacity: specialty-aware experts may be naturally selected for clinical-extraction contexts. The implication for practitioners is operational rather than theoretical: the cost of using Qwen 3 235B was USD 0.21 across the five-review benchmark, less than 1.5× the GPT-4o-mini cost of USD 0.14 and approximately 5× less than the most expensive option (DeepSeek V3 at USD 1.15) for equivalent accuracy.

5.3.2. Schema design tensions

The `n_placebo` field is a misnomer for trials with active comparators (r03, r04, r05). The `n_active` field is ambiguous in multi-arm trials when the SLR authors designate one arm as "active". A revised schema — `n_comparator`, `comparator_type` ∈ {placebo, active, lower-dose, none}, plus an `arms` list with arm-level *n* and intervention — would likely improve apparent accuracy by 5–15 percentage points in r03 and r05. We deliberately did not adopt that revision because the present study's purpose was to evaluate LLMs against a faithful reproduction of the source SLRs' schema; future work should pilot the revised schema and report the gain attributable to schema redesign versus model improvement.

5.3.3. Ground-truth framing as evaluation confound

The five framing examples in Section 4.4 are *not* extraction errors: they are correct readings of the source paper that disagree with the SLR's downstream re-coding. Three recommendations follow. First, ground-truth construction protocols should annotate framing-dependent cells. Second, evaluations should report two accuracies: against the SLR's coded values and against the source paper's stated values. Third, future LLM-extraction agents should be aware of framing decisions: a system that ingests the SLR protocol along with the trial paper could in principle reproduce the SLR authors' coding; a system that ingests only the trial paper cannot.

5.4. Practical recommendations for production deployment

For practitioners considering production deployment of an LLM-extraction pipeline, we offer the following:

- **Default model: Qwen 3 235B-A22B.** Highest mean accuracy in this study (80.9%) at USD 0.21 across the five-review benchmark. Lowest SD across reviews (6.2 pp).
- **If reliability/SD is paramount: DeepSeek V3.** Equal mean accuracy (80.8%) and modest SD (6.6 pp) but at 5× the cost.
- **If cost is paramount: GPT-4o-mini.** Cheapest at USD 0.14; five percentage points below Qwen 3 235B; the multi-arm `n_active` confusion (Finding 3) is a known weakness in this model class.
- **Always apply v2 post-processing** for the `first_author` field via PubMed metadata injection. Raw LLM accuracy on this field is 0–57% across models; the v2 fix is a single-line metadata replacement that costs nothing.
- **Always run a second-pass dedup** with NCT registration numbers if available; in r02 we lost 2 ground-truth papers to dedup mis-classification (DEFINE-HF and DAPA-VO2 confused with similarly-named SGLT2 sub-studies).

5.5. Generalisability beyond medical domains

The pipeline architecture is domain-agnostic: it requires only a search-engine query, an inclusion-criteria prompt and an extraction schema. The schema, however, encodes assumptions specific to medical RCTs (n_active, n_placebo, dose). Applying the pipeline to non-medical domains (e.g., software engineering systematic reviews, social-science quantitative meta-analyses) would require schema redesign but no change to the agent topology.

6. Limitations

This study has several limitations.

Single-database retrieval. The retrieval agent queried PubMed only. The five source SLRs each searched 2–8 databases (PubMed, Embase, Cochrane CENTRAL, ClinicalTrials.gov, Web of Science, Scopus, PsycINFO, ANZCTR). Reproducing only the PubMed component of these searches is the chief reason the retrieval recall is 88.9–100% rather than a clean 100%: the two ground-truth papers we missed (Ibrahim 2020 and Papi 2017) would in principle be retrievable from non-PubMed indices.

Locked prompt. The extraction prompt was locked at version 3 on 2026-04-28 after r02 was processed; r03, r04 and r05 served as held-out validation. The decision was deliberate (to avoid prompt overfitting) but means that documented weaknesses — crossover recognition (Finding 2), multi-arm `n_active` (Finding 3), dose format normalisation — were not patched. A version 4 prompt with explicit crossover guidance would likely improve r04 `n_active` by approximately 4 pp and r05 `n_active` substantially above the 0% baseline.

Schema misnomers. The `n_placebo` field is the most visible schema problem; `n_active` in multi-arm trials is the second. We document the impact (Section 5) but did not implement the corrective schema in this study.

Single-shot inference. Each model-paper extraction was a single deterministic call (temperature 0). Self-consistency sampling (Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery and Zhou, 2023b) or model ensembling could recover some of the documented errors but at proportional cost.

English-language only. All five source SLRs and all 62 included trials are English-language. Khraisha et al. (2024) reports notable LLM accuracy degradation on non-English literature; our findings should not be over-generalised across languages.

Five medical specialties. The five domains span pediatric neurology, cardiology, oncology, psychiatry and respiratory medicine. They do not cover, for example, infectious-disease meta-analyses, diagnostic-test reviews or qualitative reviews. The cumulative-finding patterns (especially Finding 4) may not generalise to those review types.

No human-comparison study. We benchmarked the pipeline against the published SLR ground truths but did not run a head-to-head LLM-versus-human extraction study with timing and inter-rater agreement metrics. Such a study is the natural follow-up.

Single search-strategy reconstruction. For each review we reconstructed a single PubMed query from the SLR's supplementary materials. Sensitivity analyses on alternative query phrasings might shift the retrieval-recall numbers by a few percentage points.

7. Conclusion

This study provides the first cross-domain head-to-head comparison of four frontier LLMs on a multi-agent SLR pipeline applied to five published medical systematic reviews and 62 ground-truth RCTs. Two open mixture-of-experts models (Qwen 3 235B-A22B and DeepSeek V3) achieved 80.9% and 80.8% post-processed extraction accuracy respectively, exceeding the closed baseline (GPT-4o-mini at 75.9%) and the dense open model (Llama 3.3 70B at 75.1%). Mean retrieval recall across the five reviews was 96.2%, and the end-to-end cost across all four models and all five reviews was under USD 4.

The contribution of this paper is empirical and methodological. The empirical contribution is a public benchmark across five medical specialties demonstrating that open MoE models match or exceed closed baselines on structured medical extraction, and identifying four cumulative cross-review patterns that no single-domain study could have surfaced. The methodological contribution is the two-pass evaluation that separates LLM extraction failure from format-mapping failure, and the explicit documentation of ground-truth framing decisions as an evaluation confound that future SLR-automation benchmarks should acknowledge.

Three avenues for follow-up are particularly productive. First, a prompt-version-2 study that specifically targets crossover and multi-arm guidance, with the same five reviews and four models held constant, would quantify the headroom available from prompt engineering alone. Second, a head-to-head LLM-versus-human extraction study with timing data would calibrate the hours-saved-per-review claim that motivates this entire research programme. Third, a schema-redesign study that replaces `n_placebo` with `n_comparator` + `comparator_type` would test whether the field-difficulty gradient documented here can be flattened by structural improvements that do not depend on model capability.

Author contributions

Y. Challagulla designed the multi-agent pipeline, implemented the retrieval, screening, full-text, deduplication and extraction agents, constructed the per-review ground truths, ran the four-LLM evaluation across all five reviews, performed the statistical and per-field analysis, generated the figures and tables and drafted the manuscript.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All code, prompts, ground-truth files and per-paper model outputs are available in the project's public repository: https://github.com/bhaskargurram-ai/agentic-slr.

References

Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al., 2023. GPT-4 technical report. arXiv preprint arXiv:2303.08774 doi:10.48550/arXiv.2303.08774.

Ali, A.E., Mazroua, M.S., ElSaban, M., Najam, N., Kothari, A.S., Mansoor, T., Amal, T., Lee, J., Kashyap, R., 2023. Effect of dapagliflozin in patients with heart failure: A systematic review and meta-analysis. Global Heart 18, 45. doi:10.5334/gh.1258.

Allen, I.E., Olkin, I., 1999. Estimating time to conduct a meta-analysis from number of citations retrieved. JAMA 282, 634–635. doi:10.1001/jama.282.7.634.

Archontakis Barakakis, P., Tran, T., You, J.Y., Hernandez Romero, G.J., Gidwani, V., Martinez, F.J., Fortis, S., 2023. High versus medium dose of inhaled corticosteroid in chronic obstructive lung disease: A systematic review and meta-analysis. International Journal of Chronic Obstructive Pulmonary Disease 18, 469–482. doi:10.2147/COPD.S401736.

Bahji, A., Vazquez, G.H., Zarate, C.A., 2021. Comparative efficacy of racemic ketamine and esketamine for depression: A systematic review and meta-analysis. Journal of Affective Disorders 278, 542–555. doi:10.1016/j.jad.2020.09.071.

Bannach-Brown, A., Przybyła, P., Thomas, J., Rice, A.S.C., Ananiadou, S., Liao, J., Macleod, M.R., 2019. Machine learning algorithms for systematic review: reducing workload in a preclinical review of animal studies and reducing human screening error. Systematic Reviews 8, 23. doi:10.1186/s13643-019-0942-7.

Bastian, H., Glasziou, P., Chalmers, I., 2010. Seventy-five trials and eleven systematic reviews a day: how will we ever keep up? PLoS Medicine 7, e1000326. doi:10.1371/journal.pmed.1000326.

Borah, R., Brown, A.W., Capers, P.L., Kaiser, K.A., 2017. Analysis of the time and workforce needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. BMJ Open 7, e012545. doi:10.1136/bmjopen-2016-012545.

Chang, C.Y., Park, H., Malone, D.C., Wang, C.Y., Wilson, D.L., Yeh, Y.M., Van Boemmel-Wegmann, S., Lo-Ciganic, W.H., 2020. Immune checkpoint inhibitors and immune-related adverse events in patients with advanced melanoma: A systematic review and network meta-analysis. JAMA Network Open 3, e201611. doi:10.1001/jamanetworkopen.2020.1611.

Cock, P.J.A., Antao, T., Chang, J.T., Chapman, B.A., Cox, C.J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B., de Hoon, M.J.L., 2009. Biopython: freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics 25, 1422–1423. doi:10.1093/bioinformatics/btp163.

Cohen, A.M., Hersh, W.R., Peterson, K., Yen, P.Y., 2006. Reducing workload in systematic review preparation using automated citation classification. Journal of the American Medical Informatics Association 13, 206–219. doi:10.1197/jamia.M1929.

DeepSeek-AI, Liu, A., Feng, B., Xue, B., Wang, B., Wu, B., Lu, C., Zhao, C., Deng, C., Zhang, C., et al., 2024. DeepSeek-V3 technical report. arXiv preprint arXiv:2412.19437 doi:10.48550/arXiv.2412.19437.

Elliott, J.H., Synnot, A., Turner, T., Simmonds, M., Akl, E.A., McDonald, S., Salanti, G., Meerpohl, J., MacLehose, H., Hilton, J., et al., 2017. Living systematic review: 1. Introduction—the why, what, when, and how. Journal of Clinical Epidemiology 91, 23–30. doi:10.1016/j.jclinepi.2017.08.010.

Gartlehner, G., Kahwati, L., Hilscher, R., Thomas, I., Kugley, S., Crotty, K., Viswanathan, M., Nussbaumer-Streit, B., Booth, G., Erskine, N., Konet, A., Chew, R., 2024. Data extraction for evidence synthesis using a large language model: A proof-of-concept study. Research Synthesis Methods 15, 576–589. doi:10.1002/jrsm.1710.

Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., et al., 2024. The Llama 3 herd of models. arXiv preprint arXiv:2407.21783 doi:10.48550/arXiv.2407.21783.

Guo, E., Gupta, M., Deng, J., Park, Y.J., Paget, M., Naugler, C., 2024. Automated paper screening for clinical reviews using large language models: Data analysis study. Journal of Medical Internet Research 26, e48996. doi:10.2196/48996.

Hamel, C., Hersi, M., Kelly, S.E., Tricco, A.C., Straus, S., Wells, G., Pham, B., Hutton, B., 2021. Guidance for using artificial intelligence for title and abstract screening while conducting knowledge syntheses. BMC Medical Research Methodology 21, 285. doi:10.1186/s12874-021-01451-2.

Harrison, H., Griffin, S.J., Kuhn, I., Usher-Smith, J.A., 2020. Software tools to support title and abstract screening for systematic reviews in healthcare: an evaluation. BMC Medical Research Methodology 20, 7. doi:10.1186/s12874-020-0897-3.

Higgins, J.P.T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M.J., Welch, V.A., 2019. Cochrane Handbook for Systematic Reviews of Interventions. 2 ed., John Wiley & Sons, Chichester, UK. doi:10.1002/9781119536604.

Howard, B.E., Phillips, J., Miller, K., Tandon, A., Mav, D., Shah, M.R., Holmgren, S., Pelch, K.E., Walker, V., Rooney, A.A., et al., 2016. SWIFT-Review: a text-mining workbench for systematic review. Systematic Reviews 5, 87. doi:10.1186/s13643-016-0263-z.

Khraisha, Q., Put, S., Kappenberg, J., Warraitch, A., Hadfield, K., 2024. Can large language models replace humans in systematic reviews? Evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages. Research Synthesis Methods 15, 616–626. doi:10.1002/jrsm.1715.

Larkin, J., Chiarion-Sileni, V., Gonzalez, R., Grob, J.J., Cowey, C.L., Lao, C.D., Schadendorf, D., Dummer, R., Smylie, M., Rutkowski, P., et al., 2015. Combined nivolumab and ipilimumab or monotherapy in untreated melanoma. New England Journal of Medicine 373, 23–34. doi:10.1056/NEJMoa1504030.

Lattanzi, S., Trinka, E., Russo, E., Del Giovane, C., Matricardi, S., Meletti, S., Striano, P., Tabaee Damavandi, P., Silvestrini, M., Brigo, F., 2023. Pharmacotherapy for dravet syndrome: A systematic review and network meta-analysis of randomized controlled trials. Drugs 83, 1409–1424. doi:10.1007/s40265-023-01936-y.

Lee, J., Cha, W., Park, Y., Cha, W.C., 2024. Evaluation methods for large language models in medical field: A systematic review. BMC Medical Informatics and Decision Making 24, 248. doi:10.1186/s12911-024-02709-7.

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., et al., 2023. Holistic evaluation of language models. Transactions on Machine Learning Research doi:10.48550/arXiv.2211.09110.

Maity, S., Saikia, M.J., 2025. Large language models in healthcare and medical applications: A review. Bioengineering 12, 631. doi:10.3390/bioengineering12060631.

Marshall, I.J., Kuiper, J., Banner, E., Wallace, B.C., 2017. Automating biomedical evidence synthesis: RobotReviewer. Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics-System Demonstrations , 7–12doi:10.18653/v1/P17-4002.

Marshall, I.J., Wallace, B.C., 2019. Toward systematic review automation: a practical guide to using machine learning tools in research synthesis. Systematic Reviews 8, 163. doi:10.1186/s13643-019-1074-9.

McMurray, J.J.V., Solomon, S.D., Inzucchi, S.E., Køber, L., Kosiborod, M.N., Martinez, F.A., Ponikowski, P., Sabatine, M.S., Anand, I.S., Bělohlávek, J., et al., 2019. Dapagliflozin in patients with heart failure and reduced ejection fraction. New England Journal of Medicine 381, 1995–2008. doi:10.1056/NEJMoa1911303.

OpenAI, 2024. OpenAI API documentation. https://platform.openai.com/docs/. Accessed: 2026-04-29.

Ouzzani, M., Hammady, H., Fedorowicz, Z., Elmagarmid, A., 2016. Rayyan—a web and mobile app for systematic reviews. Systematic Reviews 5, 210. doi:10.1186/s13643-016-0384-4.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., et al., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372, n71. doi:10.1136/bmj.n71.

Popova, V., Daly, E.J., Trivedi, M., Cooper, K., Lane, R., Lim, P., Mazzucco, C., Hough, D., Thase, M.E., Shelton, R.C., et al., 2019. Efficacy and safety of flexibly dosed esketamine nasal spray combined with a newly initiated oral antidepressant in treatment-resistant depression: A randomized double-blind active-controlled study. American Journal of Psychiatry 176, 428–438. doi:10.1176/appi.ajp.2019.19020172.

Pressman, S.M., Borna, S., Gomez-Cabello, C.A., Haider, S.A., Forte, A.J., 2024. Clinical and surgical applications of large language models: A systematic review. Journal of Clinical Medicine 13, 3041. doi:10.3390/jcm13113041.

Przybyła, P., Brockmeier, A.J., Kontonatsios, G., Le Pogam, M.A., McNaught, J., von Elm, E., Nolan, K., Ananiadou, S., 2018. Prioritising references for systematic reviews with RobotAnalyst: A user study. Research Synthesis Methods 9, 470–488. doi:10.1002/jrsm.1311.

Pydantic Team, 2024. Pydantic: Data validation using python type hints. https://docs.pydantic.dev/. Accessed: 2026-04-29.

Rabe, K.F., Martinez, F.J., Ferguson, G.T., Wang, C., Singh, D., Wedzicha, J.A., Trivedi, R., St. Rose, E., Ballal, S., McLaren, J., et al., 2020. Triple inhaled therapy at two glucocorticoid doses in moderate-to-very-severe COPD. New England Journal of Medicine 383, 35–48. doi:10.1056/NEJMoa1916046.

Sayers, E.W., Bolton, E.E., Brister, J.R., Canese, K., Chan, J., Comeau, D.C., Connor, R., Funk, K., Kelly, C., Kim, S., et al., 2022. Database resources of the National Center for Biotechnology Information. Nucleic Acids Research 50, D20–D26. doi:10.1093/nar/gkab1112.

van de Schoot, R., de Bruin, J., Schram, R., Zahedi, P., de Boer, J., Weijdema, F., Kramer, B., Huijts, M., Hoogerwerf, M., Ferdinands, G., et al., 2021. An open source machine learning framework for efficient and transparent systematic reviews. Nature Machine Intelligence 3, 125–133. doi:10.1038/s42256-020-00287-7.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q.V., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: International Conference on Learning Representations.

Srivastava, A., Rastogi, A., Rao, A., Shoeb, A.A.M., Abid, A., Fisch, A., Brown, A.R., Santoro, A., Gupta, A., Garriga-Alonso, A., et al., 2023. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. Transactions on Machine Learning Research doi:10.48550/arXiv.2206.04615.

Sundaram, G., Berleant, D., 2023. Automating systematic literature reviews with natural language processing and text mining: a systematic literature review , 73–92doi:10.1007/978-981-99-3243-6_7.

Syriani, E., David, I., Kumar, G., 2024. Screening articles for systematic reviews with ChatGPT. Journal of Computer Languages 80, 101287. doi:10.1016/j.cola.2024.101287.

Thirunavukarasu, A.J., Ting, D.S.J., Elangovan, K., Gutierrez, L., Tan, T.F., Ting, D.S.W., 2023. Large language models in medicine. Nature Medicine 29, 1930–1940. doi:10.1038/s41591-023-02448-8.

Thomas, J., Brunton, J., Graziosi, S., 2017. EPPI-Reviewer 4: software for research synthesis. EPPI-Centre Software, Social Science Research Unit, UCL Institute of Education .

Together AI, 2024. Together inference engine: Hosted open-source large language models. https://docs.together.ai/. Accessed: 2026-04-29.

Wallace, B.C., Trikalinos, T.A., Lau, J., Brodley, C., Schmid, C.H., 2010. Semi-automated screening of biomedical citations for systematic reviews. BMC Medical Informatics and Decision Making 10, 55. doi:10.1186/1472-6947-10-55.

Wang, S., Scells, H., Koopman, B., Zuccon, G., 2023a. Can ChatGPT write a good Boolean query for systematic review literature search? Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval , 1426–1436doi:10.1145/3539618.3591703.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., Zhou, D., 2023b. Self-consistency improves chain of thought reasoning in language models. International Conference on Learning Representations .

Whitton, J., Hunter, A., 2023. Automated tabulation of clinical trial results: A joint entity and relation extraction approach with transformer-based language representations. Artificial Intelligence in Medicine 138, 102510. doi:10.1016/j.artmed.2023.102510.

Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., Zhou, F., Huang, F., et al., 2025. Qwen3 technical report. arXiv preprint arXiv:2505.09388 doi:10.48550/arXiv.2505.09388.

A. Locked v3 extraction prompt

The prompt below was locked on 2026-04-28 after revision based on r01 and r02 evidence. It was applied unchanged to r03, r04 and r05.

```
You are a data extraction specialist for systematic reviews of clinical trials.

Given the full text of a randomized controlled trial, extract structured study characteristics following
PRISMA / Cochrane data extraction standards.

Rules:
1. Extract only information explicitly stated in the paper. Do not infer.
2. If a field cannot be determined from the text, use 'not_reported' for risk of bias fields and empty/zero/null
for others as appropriate.
3. For 'n_active', sum the sample sizes of ALL active treatment arms if the trial has multiple doses or multiple
active drugs.
4. For 'intervention', give only the generic drug name (e.g., 'Cannabidiol', 'Fenfluramine hydrochloride',
'Stiripentol', 'Soticlestat').
5. For age_range_years: explicit range as "2-18"; lower bound only as "≥18"; upper bound only as "≤12"; both
bounds as "X-Y"; none stated as "not_reported".
6. For maintenance_weeks_ge_12 (boolean): True if treatment or follow-up duration ≥12 weeks; the paper need
not use the word "maintenance".
7. For risk of bias, apply the Cochrane Risk of Bias tool criteria: 'low' / 'unclear' / 'high' / 'not_reported'.
```

```
    Return the extraction as JSON matching the specified schema.
```

B. Full per-field × per-review × per-model accuracy matrix

Table 8 reports raw (pre-v2) accuracy for every combination of field, review and model. Together with the v2 numbers in Table 6, this table allows readers to reconstruct any aggregate statistic reported in the paper.

C. Source SLRs and ground-truth trial PMIDs

The 62 ground-truth RCT PMIDs across the five reviews are listed below.

r01 Lattanzi 2023 (Dravet syndrome, 7 RCTs): 11089822, 29540584, 28538134, plus four additional PMIDs documented in `reviews/r01_lattanzi_dravet/ground_truth.json`.

r02 Ali 2023 (dapagliflozin in HF, 9 RCTs): 33426003, 31535829, 34446370, plus six additional documented in the project's repository.

r03 Chang 2020 (ICI in melanoma, 9 RCTs): 21639810, 28961465, 25399552, plus six additional.

r04 Bahji 2021 (ketamine in depression, 24 RCTs): 10686270, 12088953, 16894061, plus 21 additional.

r05 Archontakis Barakakis 2023 (high vs medium ICS in COPD, 13 RCTs): 24920884, 22334769, 24429127, plus 10 additional.

The complete set of PMIDs is preserved in each review's `ground_truth.json` file in the public repository.

Table 8
Raw per-field accuracy by review and model.

Field	Rev	GPT-4o-mini	Llama 3.3 70B	Qwen 3 235B	DeepSeek V3
first_author	r01	0%	29%	57%	29%
first_author	r02	0%	0%	29%	43%
first_author	r03	0%	11%	44%	67%
first_author	r04	0%	5%	23%	43%
first_author	r05	0%	0%	0%	18%
year	r01	100%	100%	100%	100%
year	r02	100%	100%	100%	100%
year	r03	100%	100%	100%	100%
year	r04	95%	95%	95%	95%
year	r05	100%	100%	100%	100%
phase	r01	57%	43%	86%	57%
phase	r02	71%	14%	71%	86%
phase	r03	100%	100%	100%	100%
phase	r04	86%	29%	68%	95%
phase	r05	64%	55%	91%	64%
n_total	r01	86%	86%	71%	86%
n_total	r02	100%	100%	100%	100%
n_total	r03	100%	100%	100%	100%
n_total	r04	91%	95%	91%	90%
n_total	r05	91%	91%	91%	91%
n_active	r01	71%	86%	86%	86%
n_active	r02	71%	100%	100%	100%
n_active	r03	56%	67%	78%	78%
n_active	r04	73%	90%	95%	81%
n_active	r05	0%	0%	0%	0%
n_placebo	r01	71%	71%	86%	86%
n_placebo	r02	71%	86%	86%	100%
n_placebo	r03	33%	22%	33%	67%
n_placebo	r04	36%	52%	64%	62%
n_placebo	r05	18%	9%	55%	45%
age_range_years	r01	57%	71%	71%	57%
age_range_years	r02	0%	43%	14%	14%
age_range_years	r03	44%	78%	44%	67%
age_range_years	r04	41%	62%	41%	62%
age_range_years	r05	55%	73%	55%	64%
intervention	r01–r05	100%	100%	100%	100%
dose	r01	86%	86%	86%	86%
dose	r02	86%	86%	86%	86%
dose	r03	56%	22%	33%	44%
dose	r04	59%	67%	59%	48%
dose	r05	27%	9%	18%	9%
primary_efficacy_outcome	r01	86%	86%	100%	86%
primary_efficacy_outcome	r02	100%	86%	100%	100%
primary_efficacy_outcome	r03	89%	89%	89%	89%
primary_efficacy_outcome	r04	73%	67%	77%	76%
primary_efficacy_outcome	r05	55%	55%	73%	82%
maintenance_weeks_ge_12	r01	100%	100%	100%	86%
maintenance_weeks_ge_12	r02	71%	57%	100%	86%
maintenance_weeks_ge_12	r03	56%	22%	89%	44%
maintenance_weeks_ge_12	r04	91%	95%	82%	86%
maintenance_weeks_ge_12	r05	100%	100%	100%	100%